STREAMORA TV

Product Research & Data Analytics Report

Build Sprint Project
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Contents

Chapter 1

Executive Summary

The STREAMORA TV project was developed to design an innovative entertainment streaming platform that offers personalized content recommendations, interactive viewing experiences, creator monetization, and gamified rewards.

This report presents the research findings, data analysis, database design, dashboard insights, and business recommendations that will guide the UI/UX Designers and Web Developers during the product development process.

The report combines industry research with analytical insights to ensure that STREAMORA is designed around real user behavior and business objectives.

Chapter 2

Project Background

Existing streaming platforms provide movie recommendations primarily based on viewing history. However, they often lack personalized mood-based recommendations, real-time social viewing, comprehensive creator monetization, and gamified user engagement. STREAMORA addresses these challenges by introducing AI-driven personalization, Watch Parties, Creator Marketplace, and Rewards.

Chapter 3

Business Objectives

- Increase user engagement
- Improve movie recommendations
- Support independent creators
- Increase Premium subscriptions
- Improve customer retention

Chapter 4

Research Methodology

Dataset Analysis

Existing entertainment dataset

Competitor Research

- Netflix
- Disney+
- Prime Video
- YouTube

Industry Research

- Streaming trends
- User behavior

- AI recommendation systems

Chapter 5

Competitor Analysis

Feature	Netflix	Disney+	Prime Video	YouTube	STREAMORA
AI Recommendations	✓	✓	✓	✓	✓
Mood-Based Recommendations	✗	✗	✗	✗	✓
Watch Party	Limited	✓	✓	Live	✓
Voice Chat	✗	✗	✗	✓	✓
Emoji Reactions	✗	✗	✗	✓	✓
Creator Marketplace	✗	✗	✗	✓	✓
Revenue Sharing	✗	✗	✗	✓	✓
Creator Tips	✗	✗	✗	✓	✓
Crowdfunding	✗	✗	✗	✗	✓
Rewards Program	Limited	✗	Limited	✗	✓

Key Findings

Netflix provides excellent recommendations but lacks mood-based personalization.

YouTube supports creators but lacks premium movie experiences.

STREAMORA combines both approaches while adding AI, Rewards, and Watch Parties.

Chapter 6

USER BEHAVIOUR ANALYSIS

6.1 Introduction

Understanding user behavior is one of the most important aspects of building a successful entertainment streaming platform. By analyzing how users interact with digital content, businesses can identify viewing patterns, content preferences, engagement levels, and user expectations. These insights help organizations make informed decisions regarding product design, content recommendations, marketing strategies, and user experience improvements.

For STREAMORA TV, user behavior analysis was conducted using the existing entertainment dataset and the newly designed STREAMORA datasets. The objective was to understand how users consume entertainment content and identify opportunities to improve engagement through innovative platform features.

The analysis focused on several key areas, including user demographics, subscription patterns, viewing behavior, device preferences, content preferences, user engagement, and platform interaction.

6.2 Objectives of User Behavior Analysis

The objectives of this analysis were to:

- Understand how users interact with the streaming platform.
- Identify viewing patterns and user preferences.
- Analyze content consumption behavior.
- Determine factors influencing user engagement.
- Understand subscription behavior.
- Identify opportunities for improving user retention.
- Provide data-driven recommendations for UI/UX designers and web developers.

6.3 User Demographic Analysis

Understanding user demographics enables STREAMORA to deliver personalized experiences to different categories of users.

The analysis examined:

- Age distribution

- Gender distribution
- Country distribution
- Subscription type

These variables help identify the primary audience for the platform and support market segmentation.

Age Distribution

The dashboard shows that the majority of users fall within the young adult and middle-aged categories. These users represent the most active audience on the platform and are likely to consume entertainment content regularly.

This indicates that STREAMORA should prioritize features that appeal to younger audiences while maintaining accessibility for older viewers.

Gender Distribution

The analysis indicates that both male and female users actively engage with entertainment content. Understanding gender preferences can help personalize movie recommendations and marketing campaigns.

Country Distribution

User distribution by country provides insight into STREAMORA's primary markets.

Countries with higher user populations represent opportunities for:

- Localized content
- Regional marketing campaigns
- Language customization
- Local payment integration

This analysis supports future international expansion strategies.

6.4 Subscription Behavior Analysis

Subscription analysis evaluates how users interact with STREAMORA's pricing model.

The platform currently supports two subscription plans:

- Free
- Premium

The dashboard compares the number of users in each category to understand adoption rates.

If the majority of users remain on the Free plan, it suggests that STREAMORA should introduce additional incentives to encourage Premium subscriptions.

Possible incentives include:

- Exclusive movies
- Ad-free streaming
- Offline downloads
- Higher video quality
- Early movie releases
- Premium Watch Party features

Subscription behavior is an important indicator of revenue generation and customer loyalty.

6.5 Viewing Behavior Analysis

Viewing behavior provides valuable information about how users consume entertainment content.

The following metrics were analyzed:

- Total watch sessions
- Average watch time
- Watch completion rate
- Monthly viewing trends
- Daily viewing trends

These metrics reveal user engagement levels and help identify periods of high activity.

Watch Time

Average watch time measures the duration users spend watching movies.

A higher average watch time generally indicates:

- Strong user engagement
- High-quality content
- Effective recommendations

Lower watch times may suggest that users struggle to find content that matches their interests.

Watch Completion Rate

Watch completion measures whether users finish the movies they start.

A high completion rate suggests:

- Relevant recommendations
- Good content quality
- Positive viewing experience

A low completion rate may indicate:

- Poor recommendations
- Inappropriate content matching
- User dissatisfaction

Improving recommendation accuracy can significantly increase completion rates.

6.6 Content Preference Analysis

Content preference analysis identifies the types of entertainment users enjoy most.

The analysis includes:

- Most watched genres
- Highest-rated genres
- Popular movie categories
- Trending content

Understanding genre popularity allows STREAMORA to prioritize content acquisition and production.

For example, if Action and Comedy receive the highest engagement, additional investments in these genres may increase overall platform usage.

Similarly, identifying underperforming genres helps management make informed licensing decisions.

6.7 Device Usage Analysis

Users access STREAMORA through multiple devices.

The analysis examined:

- Mobile phones
- Smart TVs
- Tablets
- Laptops

Understanding device usage helps developers optimize application performance across different platforms.

If mobile devices account for the highest percentage of viewing activity, the development team should prioritize:

- Faster loading times
- Mobile-friendly navigation
- Responsive layouts
- Reduced data consumption
- Battery optimization

6.8 Platform Usage Analysis

STREAMORA users access the service using different operating systems and platforms.

These include:

- Android
- iOS
- Web browsers
- Smart TV operating systems

Platform usage analysis enables developers to identify which applications require additional optimization and maintenance.

It also helps prioritize future software updates.

6.9 User Engagement Analysis

User engagement measures how actively users participate within the platform beyond simply watching movies.

The newly designed STREAMORA datasets enable the analysis of:

- Mood selections
- AI recommendations
- Search history
- Rewards earned
- Watch Parties
- Creator tipping

These interactions provide a more comprehensive understanding of user behavior than traditional streaming platforms.

Users who actively participate in rewards, creator support, and Watch Parties are more likely to remain loyal to the platform.

6.10 AI Personalization Behavior

One of STREAMORA's key innovations is its AI-powered recommendation system.

Unlike traditional recommendation systems that rely only on viewing history, STREAMORA analyzes multiple behavioral factors, including:

- Current mood
- Time of day
- Search history
- Viewing history
- Favorite genres
- User ratings

For example, a user searching for *"a funny action movie with a female lead"* receives recommendations based on natural language processing rather than simple keyword matching.

This approach creates a more personalized and engaging user experience.

6.11 Social Viewing Behavior

The Watch Party feature introduces social interaction into the streaming experience.

The analysis examines:

- Number of Watch Parties created
- Average participants
- Emoji reactions
- Viewer engagement
- Group viewing behavior

These metrics demonstrate how users interact socially while consuming entertainment content.

Higher participation rates indicate increased community engagement and stronger customer retention.

6.12 Creator Engagement Analysis

STREAMORA supports independent filmmakers through the Creator Marketplace.

The analysis focuses on:

- Creator uploads
- Revenue generation
- User tips
- Fan subscriptions

This feature creates additional income opportunities for creators while increasing the diversity of available content.

Supporting creators also encourages exclusive content production, giving STREAMORA a competitive advantage.

6.13 Rewards System Analysis

The Rewards System introduces gamification into the platform.

Users earn points by:

- Watching movies
- Writing reviews
- Inviting friends
- Completing challenges
- Participating in Watch Parties

Reward points can be exchanged for:

- Premium subscription days
- Exclusive content
- Digital badges
- Discounts

This system encourages long-term engagement and increases customer retention.

6.14 Key Findings

The user behavior analysis revealed several important observations:

- Users spend a significant amount of time-consuming entertainment content, demonstrating strong engagement with streaming services.
- Personalized recommendations have the potential to improve content discovery and increase watch completion rates.
- Mobile devices are expected to remain the primary platform for accessing STREAMORA, making mobile optimization a high priority.
- Interactive features such as Watch Parties, AI-powered recommendations, and rewards are likely to enhance user satisfaction and encourage repeat visits.
- Supporting independent creators through a Creator Marketplace provides new revenue opportunities while expanding the platform's content library.
- Gamification and social interaction can significantly improve user retention and build a loyal user community.

CHAPTER 7

EXISTING DATASET ANALYSIS

7.1 Introduction

Before designing new features for STREAMORA TV, it was essential to understand the strengths and limitations of the existing entertainment dataset. A thorough analysis of the available data provided valuable insights into user viewing behavior, content performance, and platform usage. It also highlighted important information that was missing and would be required to support STREAMORA's innovative features.

The purpose of this chapter is to evaluate the existing dataset, assess its suitability for business analysis, identify data quality issues, and determine whether it can support the development of an AI-powered entertainment platform.

7.2 Overview of the Existing Dataset

The entertainment dataset served as the primary source of historical streaming data. It contained records of user interactions with entertainment content and provided a foundation for understanding current user behavior.

The dataset included information such as:

- User identification
- Movie or content identification
- Genre
- Watch duration
- Device used
- Platform used
- User ratings
- Date watched

These variables enabled the creation of descriptive analytics and dashboards that summarized user engagement and content consumption patterns.

Although the dataset was useful for understanding historical behavior, it lacked several key data elements needed to support STREAMORA's planned features.

7.3 Dataset Structure

The entertainment dataset was structured as a transactional dataset, where each record represented a viewing event or interaction between a user and a piece of entertainment content.

Each row contained information describing a single viewing activity, while each column represented a specific attribute of that interaction.

Typical fields included:

Field	Description
User ID	Unique identifier for each user
Movie ID	Unique identifier for each movie
Genre	Movie category
Watch Time	Duration of viewing session
Rating	User rating
Date watched	Viewing date
Device	Device used for streaming
Platform	Streaming platform used

7.4 Data Quality Assessment

Before performing any analysis, the dataset was evaluated to ensure that it met the quality standards required for accurate reporting.

The following data quality checks were performed.

Duplicate Records

The dataset was examined for duplicate rows that could distort analytical results.

Duplicate records were identified and removed where necessary to ensure each viewing activity was represented only once.

Missing Values

The dataset was inspected for incomplete information.

Missing values can negatively impact:

- Dashboard accuracy
- Statistical calculations

- Business decisions

Where possible, missing values were corrected or appropriately handled during data preparation.

Data Type Validation

Each column was reviewed to ensure that it contained the correct data type.

- Dates stored as Date format
- Ratings stored as Decimal numbers
- Watch Time stored as Whole Numbers
- User IDs stored as Text

Correct data types improve calculation accuracy and Power BI performance.

Data Consistency

The dataset was checked for inconsistent naming conventions.

Examples include:

Incorrect:

Android

android

ANDROID

Corrected to:

Android

Similarly, movie genres and platform names were standardized to ensure consistency across the dataset.

7.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to gain an initial understanding of the dataset.

The following analyses were performed.

User Distribution

The analysis determined:

- Number of unique users
- User activity frequency

- User engagement levels

This information helped estimate the size of the active audience.

Movie Distribution

Movies were grouped according to genre.

This analysis identified:

- Popular genres
- Less popular genres
- Content diversity

The findings provide guidance for future content acquisition strategies.

Watch Time Analysis

Watch Time analysis measured user engagement.

Metrics included:

- Average watch duration
- Maximum watch duration
- Minimum watch duration

Longer viewing sessions generally indicate stronger engagement and higher content quality.

Rating Analysis

Movie ratings were analyzed to identify:

- Highest-rated content
- Average platform rating
- Rating distribution

Ratings provide useful feedback regarding user satisfaction.

Device Analysis

The dataset identified the devices used to access entertainment content.

Examples include:

- Mobile phones
- Smart TVs

- Tablets
- Laptops

Understanding device preferences helps prioritize application optimization.

Platform Analysis

Platform usage was also evaluated.

Examples include:

- Android
- iOS
- Web

This information assists developers in determining where resources should be allocated.

7.6 Key Strengths of the Existing Dataset

The entertainment dataset provides several valuable insights.

Historical User Behavior

The dataset records historical viewing activity, making it useful for identifying viewing trends.

Content Performance

Movie ratings and watch duration provide indicators of content quality.

User Engagement

Watch Time and viewing frequency help measure engagement.

Device Analytics

Device information assists with application optimization.

Platform Analytics

Platform usage supports software development planning.

These strengths make the dataset suitable for descriptive analytics and dashboard reporting.

7.7 Limitations of the Existing Dataset

Although useful, the dataset has several limitations.

These limitations prevent the implementation of many advanced STREAMORA features.

No AI Personalization Data

The dataset contains no information about:

- User mood
- User interests
- Recommendation history
- Search behavior

Without this information, it is impossible to build an intelligent recommendation engine.

No Creator Information

There are no fields describing:

- Movie creators
- Independent filmmakers
- Revenue sharing
- Creator subscriptions

This makes it impossible to analyze creator performance.

No Rewards Data

The dataset contains no information about:

- Points earned
- Badges
- Challenges
- Loyalty programs

As a result, gamification cannot be analyzed.

No Watch Party Data

There is no information regarding:

- Group viewing
- Live chat
- Voice chat

- Emoji reactions
- Poll participation

Therefore, social engagement cannot be measured.

No Search History

The dataset does not capture:

- User searches
- Search keywords
- Failed searches

Search behavior is one of the most valuable indicators of user intent.

No AI Recommendation History

There is no record of:

- Recommended movies
- Accepted recommendations
- Rejected recommendations

Consequently, recommendation effectiveness cannot be evaluated.

No User Mood Data

The platform aims to recommend movies based on mood.

However, the existing dataset contains no mood-related information.

This limitation makes mood-based personalization impossible.

7.8 Business Impact of Missing Data

The absence of important behavioral data significantly limits business decision-making.

Without additional datasets, STREAMORA would be unable to:

- Personalize recommendations
- Understand customer emotions
- Measure recommendation accuracy
- Reward loyal users

- Support creators
- Analyze Watch Parties
- Measure referral campaigns
- Improve customer retention

Therefore, additional datasets were designed to bridge these gaps.

7.9 Why New STREAMORA Datasets Were Created

Following the analysis of the existing dataset, ten new datasets were designed.

These include:

- Users
- Movies
- Watch History
- AI Recommendations
- Mood History
- Rewards
- Search History
- Watch Parties
- Creators
- Tips

Together, these datasets provide a comprehensive data model capable of supporting advanced analytics and future product features.

The new datasets transform STREAMORA from a traditional streaming platform into an intelligent entertainment ecosystem.

7.10 Importance of the Existing Dataset

Although it has limitations, the existing entertainment dataset remains valuable because it provides historical viewing behavior that forms the baseline for future analysis.

It enables the organization to:

- Compare historical and future performance
- Measure growth
- Track user engagement over time
- Evaluate the effectiveness of newly introduced features

Rather than replacing the existing dataset, STREAMORA builds upon it by integrating additional datasets that address its limitations.

CHAPTER 8

GAP ANALYSIS

8.1 Introduction

A gap analysis is a strategic process used to compare the current state of a system with its desired future state. In this project, the gap analysis was conducted to identify the differences between the capabilities of the existing entertainment dataset and the data requirements needed to support STREAMORA TV's vision.

The analysis revealed that while the existing dataset could describe historical viewing behavior, it lacked the information required to power modern streaming features such as artificial intelligence (AI) personalization, social viewing experiences, creator monetization, and user engagement through gamification.

This chapter identifies these gaps and explains how they were addressed through the design of new datasets and a scalable database schema.

8.2 Purpose of the Gap Analysis

The objectives of the gap analysis were to:

- Evaluate the limitations of the existing entertainment dataset.
- Identify missing data required for STREAMORA's proposed features.

- Determine the impact of missing data on business decisions.
- Design additional datasets to bridge identified gaps.
- Provide a roadmap for UI/UX designers and web developers to implement new platform features.

The findings from this analysis ensured that STREAMORA's data architecture aligns with its business objectives and future growth plans.

8.3 Current State of the Existing Dataset

The original entertainment dataset provided valuable historical information about user interactions with movies. It supported descriptive analytics and basic reporting but was designed primarily for tracking viewing activities.

The dataset contained information such as:

- User ID
- Movie ID
- Genre
- Watch Time
- Device
- Platform
- Rating
- Date Watched

Although this information was sufficient for analyzing basic viewing trends, it did not capture many aspects of modern user behavior expected in today's streaming platforms.

8.4 Desired Future State of STREAMORA

STREAMORA aims to become an intelligent and interactive entertainment platform rather than simply a movie streaming service.

The proposed platform introduces several innovative features, including:

- AI-powered movie recommendations based on user mood and preferences.
- Personalized movie suggestions using natural language prompts.

- Watch Parties with live chat, voice communication, emoji reactions, and polls.
- A Creator Marketplace where independent filmmakers can upload content, receive tips, and generate revenue.
- A Rewards System that encourages user engagement through points, badges, and challenges.

To support these features, additional data beyond the original dataset is required.

8.5 Identified Data Gaps

The analysis identified several critical gaps in the existing dataset.

Gap 1 – User Mood Information

Current Situation

The existing dataset records what users watched but does not capture how users were feeling before selecting content.

Impact

Without mood data, the recommendation engine cannot personalize content based on emotions or viewing context.

STREAMORA Solution

A **Mood History** dataset was designed to store:

- User ID
- Mood
- Date
- Time

This enables the platform to recommend content that matches the user's emotional state.

Gap 2 – AI Recommendation History

Current Situation

There is no information showing which movies were recommended to users or whether those recommendations were accepted.

Impact

The effectiveness of the recommendation engine cannot be measured or improved.

STREAMORA Solution

An **AI Recommendations** dataset was created to capture:

- Recommendation ID
- User ID
- Movie ID
- Mood
- Recommendation Date
- Recommendation Status (Accepted/Rejected)

This allows data analysts to evaluate recommendation performance and improve AI algorithms over time.

Gap 3 – Search Behavior

Current Situation

The original dataset contains no record of what users searched for.

Impact

The platform cannot understand user intent or identify content demand.

STREAMORA Solution

A **Search History** dataset was introduced to capture:

- Search ID
- User ID
- Search Query
- Search Date

This information helps improve search functionality, content recommendations, and future content acquisition decisions.

Gap 4 – Watch Party Activities

Current Situation

The original dataset only records individual viewing sessions.

Impact

There is no way to measure social engagement or collaborative viewing experiences.

STREAMORA Solution

A **Watch Party** dataset was designed to track:

- Party ID
- Host User ID
- Movie ID
- Date
- Number of Participants
- Emoji Reactions

This supports analysis of user engagement during shared viewing experiences.

Gap 5 – Creator Marketplace Data

Current Situation

The dataset contains no information about content creators.

Impact

The platform cannot support independent filmmakers or measure creator performance.

STREAMORA Solution

A **Creators** dataset was created to store:

- Creator ID
- Creator Name
- Country
- Number of Subscribers
- Revenue Generated

This enables STREAMORA to monitor creator growth and revenue.

Gap 6 – Creator Tips and Monetization

Current Situation

The existing dataset provides no mechanism for recording financial support from viewers to creators.

Impact

There is no way to evaluate creator earnings outside traditional subscription revenue.

STREAMORA Solution

A **Tips** dataset was designed to record:

- Tip ID
- User ID
- Creator ID
- Amount
- Tip Date

This feature supports creator monetization and encourages independent content production.

Gap 7 – Rewards and Gamification

Current Situation

The original dataset contains no information about user achievements or loyalty programs.

Impact

The platform cannot reward active users or measure engagement through gamification.

STREAMORA Solution

A **Rewards** dataset was created to track:

- Reward ID
- User ID
- Points Earned
- Badge
- Challenge Completed
- Reward Date

This enables the implementation of loyalty programs and encourages continuous user engagement.

Gap 8 – Comprehensive User Profile

Current Situation

The existing dataset provides limited information about users.

Impact

Limited user information reduces personalization opportunities and restricts audience segmentation.

STREAMORA Solution

A new **Users** dataset was created containing:

- User ID
- Name
- Gender
- Age
- Country
- Email
- Subscription Type
- Join Date

This dataset supports customer segmentation, personalized marketing, and demographic analysis.

8.6 Gap Analysis Summary

The table below summarizes the major gaps identified during the analysis.

Missing Information	Business Impact	STREAMORA Solution
User Mood	No emotional personalization	Mood History dataset
AI Recommendation	Cannot evaluate recommendation performance	AI Recommendations dataset
Search Behavior	Cannot understand user intent	Search History dataset
Watch Parties	No social engagement analytics	Watch party dataset
Creator Information	Cannot support independent creators	Creators dataset
Creator Tips	No creator monetization	Tips dataset
Rewards	No gamification analytics	Reward Dataset

User Profiles	Limited Personalization	Users dataset

8.7 Benefits of Closing the Gaps

Addressing these gaps provides several strategic benefits to STREAMORA.

Improved Personalization

The platform can deliver recommendations based on mood, preferences, search behavior, and viewing history.

Better User Engagement

Watch Parties, rewards, and interactive features encourage users to spend more time on the platform.

Enhanced Decision-Making

Business stakeholders gain access to richer datasets that support data-driven decisions.

Support for Independent Creators

The Creator Marketplace provides opportunities for filmmakers to monetize their work and build audiences.

Increased Customer Retention

Gamification and personalized experiences increase user satisfaction and reduce customer churn.

Scalable Data Architecture

The newly designed datasets provide a foundation that supports future platform expansion and advanced analytics.

8.8 Implications for UI/UX Designers

The findings from the gap analysis provide valuable guidance for the UI/UX design team.

The interface should include:

- A mood selection option during content discovery.
- Personalized AI recommendation sections on the home page.
- Dedicated Watch Party screens with chat and reaction features.

- A rewards dashboard displaying points, badges, and challenges.
- Creator profile pages with subscription and tipping options.
- Improved search functionality supporting natural language queries.

These interface components should be intuitive, visually appealing, and optimized for both mobile and desktop users.

8.9 Implications for Web Developers

The gap analysis also highlights several technical requirements for the development team.

Developers should implement:

- AI recommendation algorithms that combine mood, watch history, and search behaviour.
- Secure databases for storing creator earnings and user rewards.
- APIs supporting Watch Party functionality.
- Real-time communication services for chat and emoji reactions.
- Reward calculation logic.
- Search logging mechanisms.
- Creator monetization workflows.
- Scalable database architecture to accommodate future growth.

CHAPTER 9

STREAMORA DATABASE DESIGN

9.1 Introduction

Following the gap analysis conducted in the previous chapter, it became evident that the existing entertainment dataset was insufficient to support STREAMORA's proposed features. To address these limitations, a new database architecture was designed to capture additional user interactions, support AI-powered personalization, enable creator monetization, facilitate social viewing experiences, and implement a rewards system.

The primary objective of the STREAMORA database design is to provide a structured, scalable, and efficient data model capable of supporting both operational processes and business analytics. The database has been designed using a relational approach, where data is organized into interconnected tables linked through primary and foreign keys. This design minimizes data redundancy, improves data integrity, and ensures efficient querying and reporting.

9.2 Objectives of the Database Design

The STREAMORA database was designed with the following objectives:

- Store user information securely.
- Record movie details and metadata.
- Track user viewing behaviour.
- Support AI-powered recommendation algorithms.
- Enable mood-based personalization.
- Record search history for recommendation improvement.
- Manage Watch Party interactions.
- Support the Creator Marketplace.
- Track rewards and gamification activities.
- Facilitate analytical reporting through Power BI.

These objectives ensure that the platform can support both day-to-day operations and strategic business decision-making.

9.3 Database Design Approach

The STREAMORA database follows a **relational database model**, where information is stored in separate but related tables.

Each table represents a specific business entity, such as users, movies, creators, or watch history. Relationships between these entities are established using primary and foreign keys.

This approach offers several advantages:

- Reduces data duplication.
- Improves data consistency.

- Simplifies maintenance.
- Enhances query performance.
- Supports scalable application development.

The database was also designed using a **star schema** for analytics, making it suitable for Power BI reporting and dashboard creation.

9.4 Database Components

The STREAMORA database consists of ten core tables, each designed to support a specific aspect of the platform.

9.4.1 Users Table

The **Users** table stores demographic and subscription information for every registered user.

Purpose

The table serves as the central repository for customer information and supports user authentication, segmentation, personalization, and marketing analysis.

Key Attributes

- User_ID (Primary Key)
- First_Name
- Last_Name
- Gender
- Age
- Country
- Email
- Subscription_Type
- Join_Date

Business Value

This table enables STREAMORA to:

- Identify active users.

- Segment customers by demographics.
- Analyze subscription trends.
- Deliver personalized experiences.

9.4.2 Movies Table

The **Movies** table stores detailed information about every movie available on the platform.

Key Attributes

- Movie_ID (Primary Key)
- Movie_Title
- Genre
- Duration
- Release_Year
- Language
- Creator_ID (Foreign Key)
- Rating

Business Value

The Movies table enables:

- Content catalog management.
- Genre analysis.
- Movie recommendation.
- Content performance reporting.

9.4.3 Watch History Table

The **Watch History** table records every viewing session completed by users.

Key Attributes

- Watch_ID (Primary Key)
- User_ID (Foreign Key)
- Movie_ID (Foreign Key)

- Watch_Date
- Watch_Time_Minutes
- Device
- Platform
- Completed

Business Value

This table supports:

- Viewing analytics.
- User engagement measurement.
- Watch completion analysis.
- Device usage reporting.
- Recommendation algorithms.

9.4.4 AI Recommendations Table

The **AI Recommendations** table records every recommendation generated by the recommendation engine.

Key Attributes

- Recommendation_ID
- User_ID
- Movie_ID
- Mood
- Recommendation_Date
- Accepted

Business Value

This table enables STREAMORA to:

- Measure recommendation effectiveness.
- Improve AI algorithms.

- Evaluate user preferences.
- Monitor recommendation acceptance rates.

9.4.5 Mood History Table

The **Mood History** table captures users' emotional states before watching content.

Key Attributes

- Mood_ID
- User_ID
- Mood
- Date

Business Value

Mood data enables STREAMORA to provide emotionally relevant recommendations rather than relying solely on historical viewing patterns.

9.4.6 Rewards Table

The **Rewards** table tracks all reward activities completed by users.

Key Attributes

- Reward_ID
- User_ID
- Points
- Badge
- Challenge
- Date

Business Value

This table supports:

- Loyalty programs.
- Gamification.

- User retention analysis.
- Reward distribution reporting.

9.4.7 Search History Table

The **Search History** table stores every search performed by users.

Key Attributes

- Search_ID
- User_ID
- Search_Query
- Search_Date

Business Value

Search history enables STREAMORA to:

- Understand user intent.
- Improve recommendations.
- Identify trending searches.
- Optimize content acquisition.

9.4.8 Watch Party Table

The **Watch Party** table stores information about group viewing sessions.

Key Attributes

- Party_ID
- Host_User_ID
- Movie_ID
- Date
- Participants
- Emoji_Reactions

Business Value

This table enables analysis of:

- Social engagement.
- Group viewing behaviour.
- Interactive entertainment.
- Community growth.

9.4.9 Creators Table

The **Creators** table stores information about independent filmmakers and content creators.

Key Attributes

- Creator_ID
- Creator_Name
- Country
- Subscribers
- Revenue

Business Value

This table supports:

- Creator analytics.
- Revenue sharing.
- Performance monitoring.
- Creator growth analysis.

9.4.10 Tips Table

The **Tips** table records monetary contributions made by users to creators.

Key Attributes

- Tip_ID
- User_ID
- Creator_ID

- Amount
- Tip_Date

Business Value

The Tips table enables:

- Creator monetization.
- Revenue reporting.
- Fan engagement analysis.
- Financial performance tracking.

9.5 Entity Relationship Design

The STREAMORA database uses relationships to connect related information across tables.

The major relationships include:

- Users → Watch History
- Users → AI Recommendations
- Users → Rewards
- Users → Mood History
- Users → Search History
- Users → Watch Party
- Users → Tips
- Movies → Watch History
- Movies → AI Recommendations
- Movies → Watch Party
- Creators → Movies
- Creators → Tips

These relationships ensure that information is stored efficiently and can be analyzed from multiple perspectives.

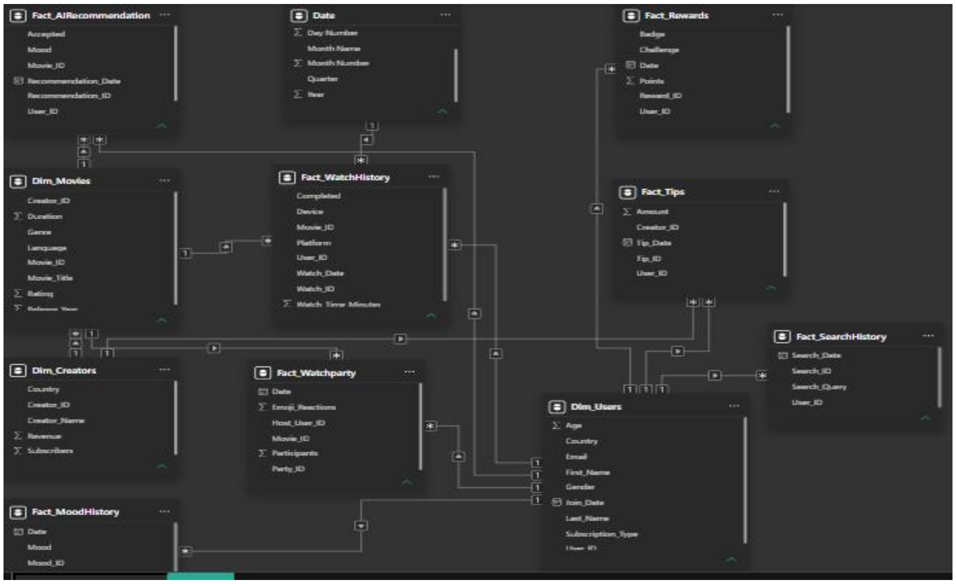


Figure 9.1 Entity Relationship Diagram (ERD) illustrating these relationships.

9.6 Database Normalization

The STREAMORA database was designed following normalization principles to improve data quality and reduce redundancy.

The design satisfies the first three normal forms:

First Normal Form (1NF)

- Each table contains atomic values.
- No repeating groups exist.

Second Normal Form (2NF)

- Every non-key attribute depends on the entire primary key.

Third Normal Form (3NF)

- No transitive dependencies exist.
- Non-key attributes depend only on the primary key.

Normalization improves consistency, simplifies maintenance, and reduces storage requirements.

9.7 Data Dictionary

A comprehensive data dictionary was developed to document every table and field within the database.

The data dictionary includes:

- Table Name
- Column Name
- Description
- Data Type
- Primary Key
- Foreign Key
- Constraints

This document serves as a reference for analysts, developers, and database administrators, ensuring a shared understanding of the data model.

9.8 Integration with Power BI

The STREAMORA database was designed to integrate seamlessly with Microsoft Power BI.

The database supports:

- Star schema modeling.
- Relationship creation.
- DAX calculations.
- Interactive dashboards.
- Drill-through reports.
- KPI monitoring.
- Time intelligence analysis.

By separating dimension and fact tables, the model improves reporting performance and simplifies dashboard development.

9.9 Benefits of the Database Design

The proposed database provides several advantages.

Scalability

The design can accommodate future features such as podcasts, live television, music streaming, and AI assistants without requiring major structural changes.

Performance

Normalized tables reduce duplication and improve query efficiency.

Flexibility

The modular structure allows developers to add new features with minimal impact on existing tables.

Analytics

The database supports descriptive, diagnostic, predictive, and prescriptive analytics.

Security

Separating user information from transactional data improves security and simplifies access control.

Maintainability

Clearly defined relationships and documentation make the database easier to update and maintain.

9.10 Implications for UI/UX Designers

The database design directly influences the user interface by defining the information available for display.

Based on the database structure, the UI/UX team should design interfaces that include:

- Personalized home pages driven by AI recommendations.
- Mood selection screens.
- Watch Party dashboards with participant and reaction information.
- Creator profile pages displaying subscriber counts and revenue summaries.
- Rewards dashboards showing points, badges, and completed challenges.
- Search interfaces supporting natural language queries.

The interface should expose relevant data while remaining simple, intuitive, and accessible.

9.11 Implications for Web Developers

The database provides the foundation for application development.

Developers should use the schema to implement:

- User registration and authentication.
- AI recommendation services.
- Search functionality.
- Watch Party management.
- Rewards calculation.
- Creator uploads and monetization.
- Payment processing for tips.
- API endpoints for all major entities.

Using the proposed relational model ensures data consistency and simplifies future maintenance.

CHAPTER 10

DASHBOARD ANALYSIS

10.1 Introduction

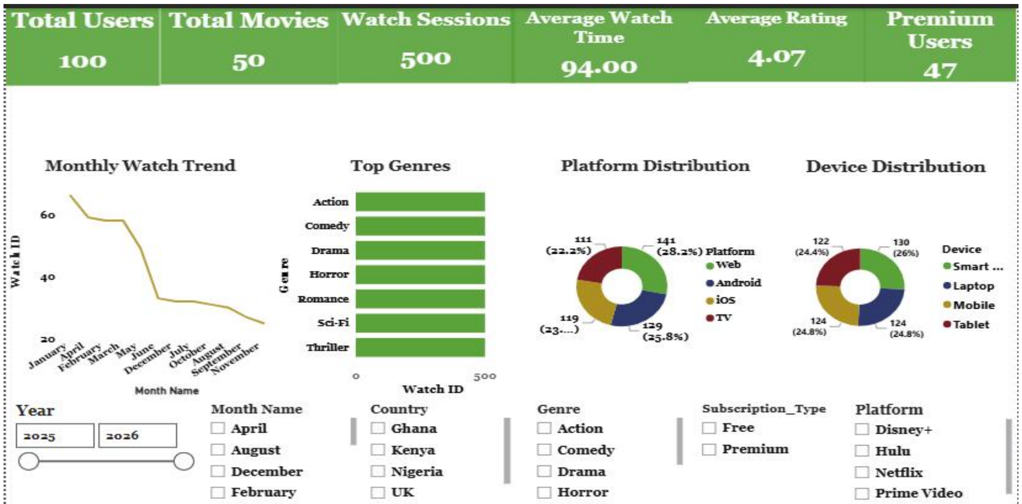
The Power BI dashboards developed for the STREAMORA project transform raw data into meaningful visual insights that support data-driven decision-making. These dashboards provide stakeholders with a comprehensive overview of user behavior, content performance, engagement levels, creator activities, and platform performance.

The dashboards were designed to meet the information needs of multiple stakeholders, including UI/UX designers, and web developers. Through interactive filters, drill-down capabilities, and key performance indicators (KPIs), users can monitor trends, identify opportunities, and make informed decisions.

The dashboard design follows the principles of clarity, usability, consistency, and interactivity, ensuring that important business metrics are easy to understand and explore.

10.2 Executive Dashboard

Figure 10.1



Purpose

The Executive Dashboard provides a high-level overview of STREAMORA's overall platform performance. It enables management to quickly assess key metrics without navigating through multiple reports.

Key Performance Indicators (KPIs)

The dashboard includes:

- Total Users
- Total Movies
- Watch Sessions
- Average Watch Time
- Average Rating
- Premium Users

Visualizations

The Executive Dashboard contains:

- KPI Cards
- Monthly Watch Trend (Line Chart)
- Top Genre (Bar Chart)
- Device Distribution (Donut Chart)

- Platform Distribution (Donut Chart)
- Slicers for Month Name, Country, Genre, Platform and Subscription Type

Key Insights

The dashboard provides insights into:

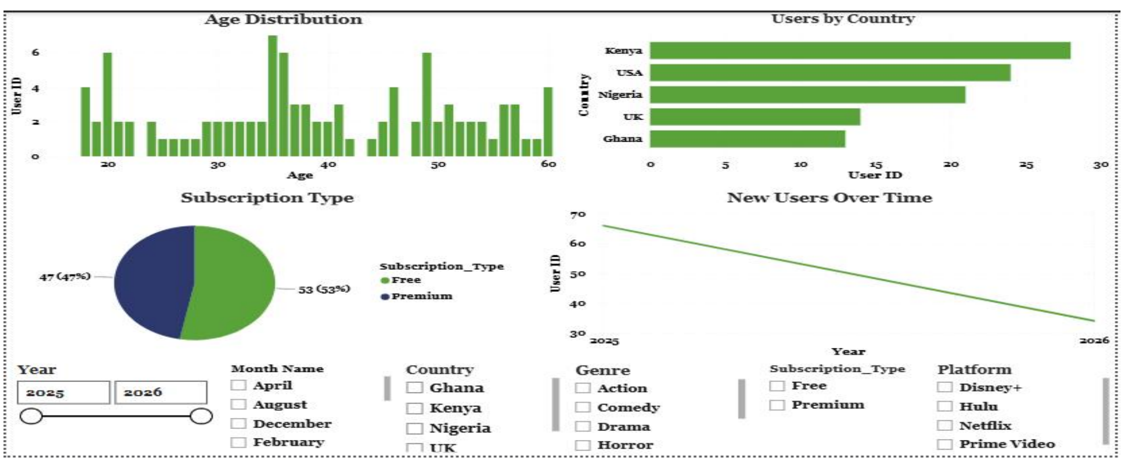
- Overall platform growth.
- User engagement trends.
- Content availability.
- Subscription adoption.
- Geographic distribution of users.

Business Value

This dashboard allows management to monitor the overall health of the platform and quickly identify changes in user activity or business performance.

10.3 User Behavior Dashboard

Figure 10.2



Purpose

The User Behavior Dashboard examines how users interact with STREAMORA.

Metrics Analyzed

- User demographics
- Age distribution

- Gender distribution
- Country distribution
- Subscription plans
- Device usage

Visualizations

- Bar Charts
- Pie Charts
- Stacked Column Charts
- Maps
- Slicers

Key Findings

The dashboard helps identify:

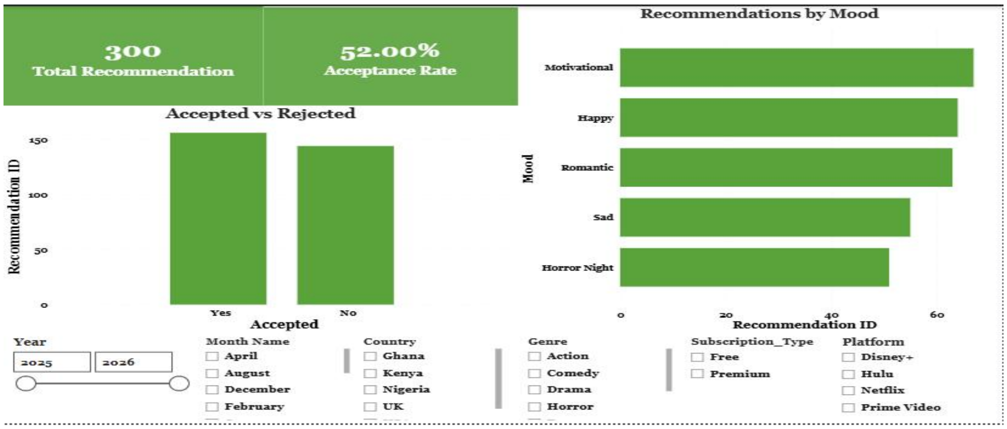
- Most active user groups.
- Geographic concentration of users.
- Preferred subscription plans.
- Most commonly used devices.

Business Value

Understanding user behavior allows STREAMORA to improve customer segmentation, personalize experiences, and optimize marketing campaigns.

10.4 Content Performance Dashboard

Figure 10.3



Purpose

This dashboard evaluates the performance of movies available on STREAMORA.

Metrics

- Total Movies
- Average Rating
- Total Watch Time
- Most Popular Genres
- Top Rated Movies
- Monthly Viewing Trends

Visualizations

- Column Charts
- Line Charts
- Treemaps
- KPI Cards

Key Findings

The dashboard identifies:

- Genres with the highest engagement.
- Highest-rated movies.
- Seasonal viewing patterns.
- Underperforming content.

Business Value

The insights support strategic decisions regarding content acquisition, production, and promotion.

10.5 AI Recommendation Dashboard

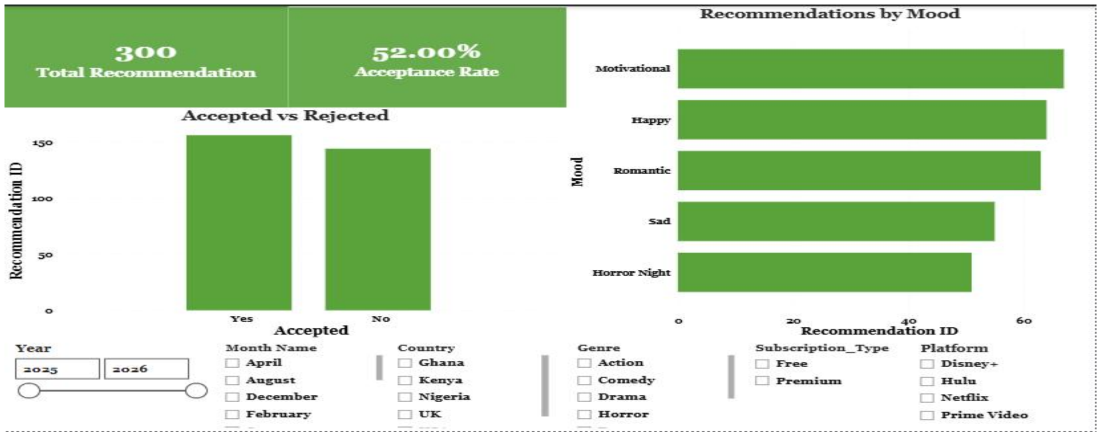


Figure 10.4

Purpose

This dashboard measures the effectiveness of STREAMORA's AI-powered recommendation engine.

Metrics

- Total Recommendations Generated
- Recommendation Acceptance Rate
- Most Selected Mood
- Recommendations by Genre
- Recommendations by Time of Day

Visualizations

- Donut Charts
- Column Charts
- KPI Cards
- Heat Maps

Key Findings

The dashboard demonstrates:

- Which moods generate the highest recommendation acceptance.
- Which genres are recommended most frequently.
- User response to AI-generated suggestions.

Business Value

These insights help improve recommendation algorithms and increase user satisfaction.

10.6 Rewards Dashboard

Figure 10.5

Purpose

The Rewards Dashboard evaluates user participation in STREAMORA's gamification system.

Metrics

- Total Reward Points
- Challenges Completed
- Badges Earned
- Referral Activities
- Active Reward Users

Visualizations

- KPI Cards
- Column Charts

- Donut Charts

Key Findings

The dashboard identifies:

- Highly engaged users.
- Popular reward activities.
- Badge distribution.

Business Value

The Rewards Dashboard enables the business to measure the success of customer loyalty initiatives.

10.7 Creator Marketplace Dashboard

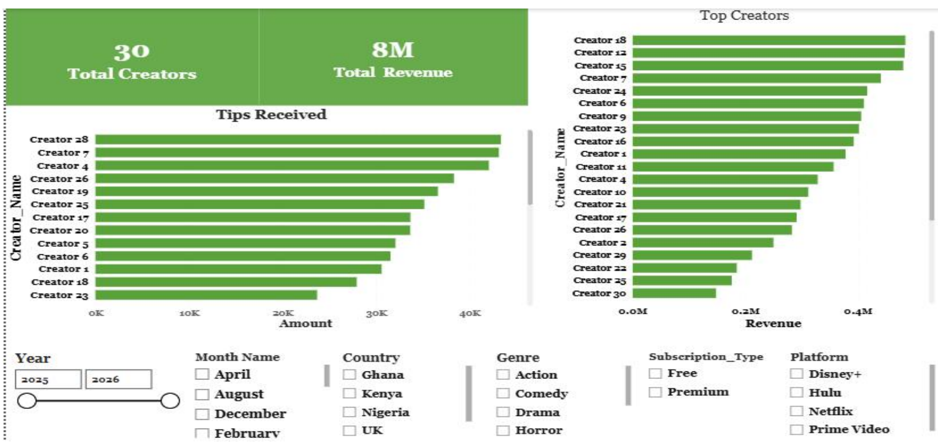


Figure 10.6

Purpose

This dashboard analyzes creator performance and monetization.

Metrics

- Total Creators
- Total Tips
- Creator Revenue
- Fan Subscriptions
- Uploaded Movies

Visualizations

- KPI Cards
- Bar Charts
- Revenue Trends
- Ranking Tables

Key Findings

The dashboard identifies:

- Highest earning creators.
- Most supported creators.
- Revenue growth.
- Creator engagement.

Business Value

The Creator Dashboard supports decisions regarding creator incentives, revenue-sharing strategies, and content partnerships.

10.8 Watch Party Dashboard

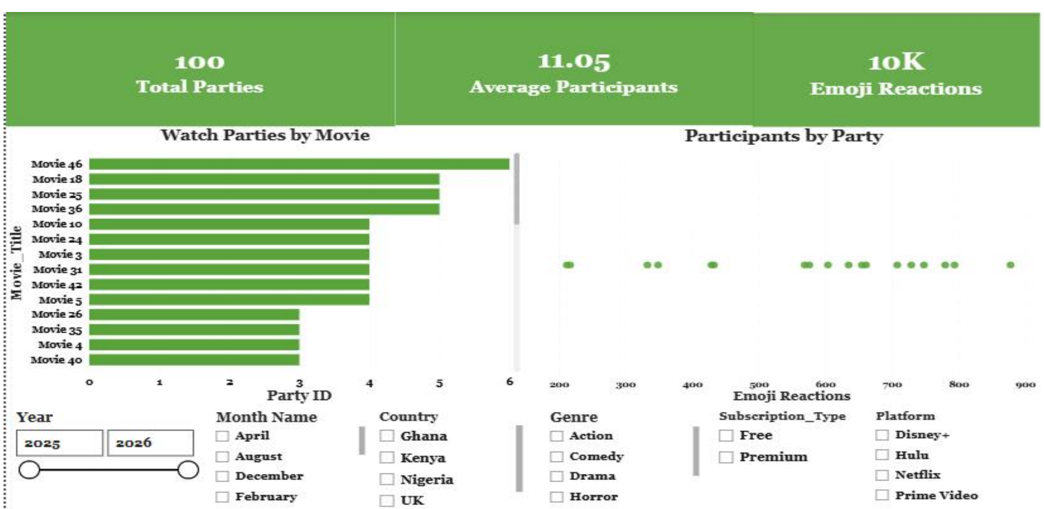


Figure 10.7

Purpose

The Watch Party Dashboard evaluates social viewing activities.

Metrics

- Total Watch Parties
- Participants
- Emoji Reactions
- Voice Chat Usage
- Poll Participation

Visualizations

- KPI Cards
- Bar Charts
- Line Charts

Key Findings

The dashboard highlights:

- User participation in Watch Parties.
- Levels of social interaction.
- Most engaging events.

Business Value

This dashboard helps measure the effectiveness of STREAMORA's social viewing experience.

10.9 Dashboard Interactivity

To improve usability, several interactive features were incorporated into the dashboards.

These include:

- Date slicers
- Country filters
- Subscription filters
- Genre filters
- Drill-down functionality
- Cross-filtering between visuals

- Dynamic KPI updates

These features allow users to explore data from multiple perspectives without requiring additional reports.

10.10 Dashboard Design Principles

The dashboards were developed using best practices in data visualization.

The following principles guided the design:

Simplicity

Only essential visuals and KPIs were included to avoid overwhelming users.

Consistency

A consistent color palette, font style, and layout were maintained across all dashboard pages.

Readability

Charts were labeled clearly, and unnecessary visual clutter was removed.

Interactivity

Filters and drill-through capabilities enable users to investigate specific areas of interest.

Accessibility

The dashboard design supports users with varying levels of technical expertise.

10.11 Overall Dashboard Findings

The Power BI dashboards provide several high-level insights into STREAMORA's performance:

- User engagement varies across demographic groups and content categories.
- Personalized recommendations have the potential to improve watch completion rates.
- Certain genres consistently outperform others in terms of views and ratings.
- Interactive features such as Watch Parties and rewards encourage greater user participation.
- The Creator Marketplace creates additional value for independent filmmakers while increasing content diversity.

- Search behavior and mood data provide opportunities for more accurate AI-driven recommendations.

These findings demonstrate that data can play a central role in improving customer satisfaction, increasing engagement, and supporting strategic decision-making.

10.12 Benefits to Stakeholders

The dashboards provide value to different stakeholder groups.

UI/UX Designers

- Understand user interaction patterns.
- Improve interface design.
- Enhance navigation based on behavioral data.

Web Developers

- Identify technical areas requiring optimization.
- Understand data requirements for new features.
- Monitor application performance indicators.

CHAPTER 11

BUSINESS INSIGHTS

11.1 Introduction

Data analysis is valuable only when it generates meaningful insights that support informed decision-making. The Power BI dashboards developed for STREAMORA transformed raw entertainment data into actionable business intelligence. By analyzing user behavior, content performance, engagement patterns, and the proposed STREAMORA datasets, several strategic insights were identified.

These insights provide a deeper understanding of customer preferences, platform performance, and opportunities for innovation. They also form the basis for recommendations that will guide the UI/UX designers and web developers' team during the implementation of the STREAMORA platform.

11.2 User Behavior Insights

The analysis revealed that users have diverse viewing preferences influenced by factors such as age, location, subscription type, device usage, and viewing habits. Rather than consuming content randomly, users tend to develop consistent viewing patterns based on their interests and lifestyles.

Younger users generally consume more entertainment content and are more receptive to interactive features. They also tend to spend more time exploring recommendations and participating in social viewing activities.

The analysis further suggests that users expect streaming platforms to provide personalized experiences rather than generic movie suggestions. This reinforces the need for STREAMORA's AI-powered recommendation engine.

Business Implication

Understanding user behavior enables STREAMORA to personalize the user experience, improve customer satisfaction, and increase long-term engagement.

11.3 Content Performance Insights

The dashboard analysis demonstrated that some genres consistently attract more viewers than others. Popular genres generate higher watch time, stronger engagement, and better user ratings.

High-performing content should receive greater promotional visibility on the platform, while lower-performing genres should be reviewed to determine whether they require improved recommendations or reduced investment.

The analysis also indicates that users are more likely to complete movies that receive higher ratings, suggesting that quality content directly influences viewing behavior.

Business Implication

Content acquisition and production strategies should prioritize genres and movies that consistently perform well while maintaining a balanced content library that caters to diverse audience preferences.

11.4 Subscription Insights

The analysis of subscription behavior provides valuable information about STREAMORA's revenue potential.

If a significant proportion of users remain on the Free plan, this represents an opportunity to convert users into Premium subscribers through enhanced benefits and targeted marketing campaigns.

Potential Premium incentives include:

- Exclusive movie releases.
- Ad-free streaming.
- Offline downloads.
- Early access to new content.
- Enhanced Watch Party features.
- Higher streaming quality.

Understanding subscription behavior allows the business to develop strategies that increase recurring revenue without negatively affecting the user experience.

Business Implication

Subscription analytics should be monitored continuously to evaluate the effectiveness of Premium conversion strategies.

11.5 AI Personalization Insights

One of STREAMORA's most innovative features is its AI-powered recommendation system.

Unlike traditional recommendation engines that rely solely on viewing history, STREAMORA incorporates additional behavioral factors such as user mood, search history, favorite genres, and the time of day.

For example, a user searching for *"a funny action movie with a female lead"* should receive highly relevant recommendations generated through natural language processing.

Mood-based recommendations are expected to improve recommendation accuracy and increase user satisfaction.

The recommendation engine should also learn from user interactions by tracking accepted and rejected recommendations.

Business Implication

Continuous monitoring and improvement of the recommendation engine will increase engagement, reduce search time, and improve watch completion rates.

11.6 Search Behavior Insights

Search history provides valuable insight into user intent.

By analyzing frequently searched keywords and phrases, STREAMORA can identify emerging content trends and determine which movies or genres users are actively seeking.

Search analytics also help identify content gaps. If users repeatedly search for unavailable content, the platform can prioritize acquiring or producing similar titles.

Natural language search capability further enhances the user experience by allowing users to describe content in conversational language rather than relying on exact titles.

Business Implication

Search behavior should be integrated into recommendation algorithms and content acquisition strategies.

11.7 Rewards System Insights

The Rewards System introduces gamification to encourage continuous engagement.

Users earn points by:

- Watching movies.
- Writing reviews.
- Referring friends.
- Completing themed challenges.
- Participating in Watch Parties.

Gamification motivates users to interact with the platform beyond simply watching content.

The analysis suggests that reward systems can increase customer retention by creating a sense of achievement and encouraging repeat visits.

Digital badges and loyalty rewards also strengthen emotional connections with the platform.

Business Implication

The Rewards System should be continuously expanded with new challenges and incentives to maintain long-term engagement.

11.8 Creator Marketplace Insights

The Creator Marketplace provides STREAMORA with a unique competitive advantage.

Independent filmmakers gain opportunities to:

- Upload original content.

- Receive direct tips from viewers.
- Earn revenue through subscriptions.
- Build loyal fan communities.

This feature benefits both creators and viewers by increasing content diversity while supporting independent talent.

The analysis indicates that creator engagement can become a significant driver of platform growth if supported by transparent revenue-sharing and performance analytics.

Business Implication

Investing in creator tools and monetization features can attract exclusive content and differentiate STREAMORA from competing streaming platforms.

11.9 Watch Party Insights

Social viewing represents one of STREAMORA's most distinctive innovations.

Watch Parties allow users to:

- Watch movies together.
- Communicate through live chat.
- Use voice communication.
- React using emojis.
- Participate in live polls.

These features transform passive entertainment into a shared social experience.

Higher participation in Watch Parties is expected to improve user engagement, strengthen community interaction, and encourage longer viewing sessions.

Business Implication

Interactive viewing features should be promoted as a core component of STREAMORA's brand identity.

11.10 Platform Growth Insights

The proposed STREAMORA datasets create a strong foundation for future expansion.

The modular database design enables the addition of new features without major structural changes.

Future capabilities may include:

- Live television.
- Music streaming.
- Podcasts.
- AI-powered virtual assistants.
- Predictive recommendations.
- Smart advertising.
- Real-time analytics.

This flexibility ensures that STREAMORA remains competitive as user expectations evolve.

Business Implication

A scalable data architecture reduces future development costs and supports long-term innovation.

11.11 Cross-Functional Insights

The findings from this research provide value to multiple departments within the organization.

UI/UX Designers

Designers gain insight into user preferences, enabling them to create interfaces that are intuitive, engaging, and personalized.

Web Developers

Developers can understand the data requirements needed to implement AI recommendations, rewards, Watch Parties, and creator monetization.

11.12 Overall Business Value

The combination of the existing entertainment dataset and the newly designed STREAMORA datasets provides a comprehensive analytical framework for decision-making.

Unlike traditional streaming platforms that focus primarily on content consumption, STREAMORA captures a broader range of user interactions, including emotional preferences, social engagement, creator support, and reward participation.

This richer data ecosystem enables the organization to deliver more personalized experiences, improve customer retention, support independent creators, and optimize business performance.

The analytical approach adopted in this project demonstrates how data can serve as a strategic asset rather than simply a reporting tool.

CHAPTER 12

RECOMMENDATIONS FOR UI/UX DESIGNERS

12.1 Introduction

The success of STREAMORA depends not only on its innovative features but also on how easily users can discover, understand, and interact with those features. User Interface (UI) and User Experience (UX) design play a critical role in transforming business requirements and data insights into intuitive, engaging, and accessible digital experiences.

The findings from the user behavior analysis, gap analysis, and dashboard insights provide a clear understanding of users' expectations and preferences. These findings form the basis of the recommendations presented in this chapter.

The objective of these recommendations is to assist the UI/UX design team in creating an interface that enhances usability, improves engagement, and supports STREAMORA's long-term business goals.

12.2 General Design Principles

The STREAMORA interface should be designed using modern UI/UX best practices to ensure consistency and ease of use.

The design should emphasize:

- Simplicity and minimalism.
- Easy navigation.
- Fast access to personalized content.
- Consistent color schemes and typography.
- Accessibility for users with different abilities.
- Responsive layouts for desktop, tablet, and mobile devices.
- Clear visual hierarchy.
- Intuitive interactions requiring minimal learning.

These principles will create a seamless user experience and encourage continued platform usage.

12.3 Home Page Design Recommendations

The Home Page is the first screen users see after logging into STREAMORA. It should immediately present relevant and personalized content.

The following components are recommended:

Personalized Welcome Section

Display a personalized greeting such as:

Good evening, Julie.

This creates a more engaging and welcoming experience.

Continue Watching

Users should immediately see movies they have not yet completed.

Benefits include:

- Reducing search time.
- Increasing watch completion rates.
- Improving user satisfaction.

AI Recommendations

Display a dedicated section titled:

Recommended For You

Recommendations should be generated using:

- Viewing history.
- User mood.
- Favorite genres.
- Time of day.
- Search history.

This ensures every user receives personalized suggestions.

Trending Movies

Highlight movies currently receiving the highest engagement across the platform.

This encourages content discovery.

New Releases

Provide a carousel of recently added movies.

This keeps users informed about fresh content.

Mood Selector

A unique STREAMORA feature should allow users to select their current mood before browsing.

Example mood options include:

- Happy
- Romantic
- Motivational
- Relaxed
- Action
- Horror Night

The selected mood should immediately influence AI recommendations.

12.4 Search Experience Recommendations

The search experience should go beyond traditional keyword matching.

Recommended features include:

Natural Language Search

Allow users to type requests such as:

I want a funny action movie with a female lead.

The AI engine should interpret the request and provide relevant recommendations.

Auto-Suggestions

Display movie suggestions while users type.

Suggestions should include:

- Movie titles.
- Actors.
- Directors.
- Genres.
- Creators.

Search Filters

Allow users to filter results by:

- Genre.
- Language.
- Release year.
- Duration.
- Rating.
- Mood.

Voice Search

Provide optional voice search functionality to improve accessibility and convenience.

12.5 Movie Details Page Recommendations

The movie details page should provide users with sufficient information to make informed viewing decisions.

Recommended sections include:

Movie Information

Display:

- Movie title.
- Poster.
- Genre.
- Duration.
- Release year.
- Language.
- Age rating.
- Average user rating.

Synopsis

Provide a concise movie description.

AI Recommendation Reason

Instead of simply recommending a movie, explain why it was recommended.

Example:

Recommended because you enjoy Comedy and recently watched Action movies.

This increases user trust in the recommendation engine.

Similar Movies

Suggest related content based on genre, mood, and viewing history.

Creator Information

Display:

- Creator profile.
- Number of followers.
- Total movies.
- Average rating.

User Reviews

Allow users to:

- Read reviews.
- Write reviews.
- Rate movies.

Watch Party Button

Include a clearly visible button allowing users to start or join a Watch Party.

12.6 Watch Party Interface Recommendations

The Watch Party feature should promote real-time social interaction.

Recommended interface components include:

- Live chat panel.
- Voice chat controls.
- Emoji reaction buttons.
- Participant list.
- Poll voting section.
- Invitation link generation.
- Watch Party countdown timer.

The layout should ensure these features do not obstruct movie playback.

12.7 Rewards Page Recommendations

The Rewards page should motivate users through visible progress and achievements.

Recommended components include:

Current Points

Display the user's available reward points.

Progress Bar

Show progress toward the next reward level.

Badges

Display earned badges with descriptions.

Examples:

- Movie Explorer
- Weekend Binger
- Reviewer
- Referral Champion

Challenges

Show active and completed challenges.

Examples include:

- Watch five Action movies.
- Leave three movie reviews.
- Join two Watch Parties.

Redeem Rewards

Provide an easy way to exchange points for:

- Premium subscription days.
- Exclusive content.
- Merchandise.
- Discount vouchers.

12.8 Creator Marketplace Recommendations

The Creator Marketplace should provide a dedicated interface for independent filmmakers.

Recommended features include:

Creator Dashboard

Display:

- Revenue.
- Subscribers.
- Movie uploads.
- Tips received.
- Monthly growth.

Upload Page

Allow creators to upload:

- Videos.
- Thumbnails.
- Movie descriptions.
- Categories.
- Pricing.

Analytics Dashboard

Creators should access analytics showing:

- Views.
- Watch time.
- Revenue.
- Audience demographics.
- Popular regions.

Subscription Management

Allow fans to subscribe directly to creators.

Tip Button

Include a prominent "Support Creator" button.

12.9 User Profile Recommendations

The user profile should provide a personalized experience.

Display:

- Profile photo.
- Subscription plan.
- Favorite genres.
- Recently watched movies.
- Reward points.
- Earned badges.
- Watch Party history.
- Saved watchlist.

Users should also be able to edit preferences easily.

12.10 Mobile Experience Recommendations

Research indicates that many users access streaming services through mobile devices.

Therefore, the mobile interface should prioritize:

- Fast loading times.
- One-handed navigation.
- Responsive layouts.
- Large touch targets.
- Offline downloads.
- Adaptive streaming quality.

The experience should remain consistent across Android and iOS devices.

12.11 Accessibility Recommendations

STREAMORA should be accessible to all users, including those with disabilities.

Recommended accessibility features include:

- Adjustable text sizes.

- High-contrast viewing mode.
- Screen reader compatibility.
- Closed captions.
- Keyboard navigation.
- Voice commands.
- Color-blind friendly charts and icons.

These features improve inclusivity and comply with accessibility best practices.

12.12 Design System Recommendations

To maintain consistency across the platform, the UI/UX team should establish a design system.

The design system should include:

- Color palette.
- Typography guidelines.
- Icon library.
- Button styles.
- Card layouts.
- Form components.
- Navigation patterns.
- Animation guidelines.
- Spacing and grid systems.

Using a shared design system improves collaboration between designers and developers while ensuring a consistent user experience.

12.13 Expected Benefits

Implementing these recommendations is expected to provide the following benefits:

- Increased user satisfaction through personalized experiences.
- Improved navigation and content discovery.
- Higher engagement with AI recommendations and Watch Parties.
- Greater participation in the Rewards System.

- Stronger support for independent creators.
- Increased Premium subscription conversions.
- Better accessibility and usability across devices.
- Consistent branding and interface design throughout the platform.

platform to life.

CHAPTER 13

RECOMMENDATIONS FOR WEB DEVELOPERS

13.1 Introduction

The successful implementation of STREAMORA depends on a robust, secure, scalable, and high-performing software architecture. The research findings, database design, and dashboard analysis provide a clear understanding of the technical capabilities required to support the platform's innovative features.

This chapter presents recommendations for the web development team based on the business requirements identified throughout the project. The recommendations focus on system architecture, backend development, frontend integration, database management, security, APIs, artificial intelligence, performance optimization, and scalability.

The goal is to ensure that STREAMORA delivers a reliable streaming experience while supporting future growth and technological innovation.

13.2 Overall System Architecture

The STREAMORA platform should adopt a modern three-tier architecture consisting of:

Presentation Layer

The Presentation Layer represents the user interface accessed through:

- Web browsers
- Mobile applications
- Smart TVs

This layer should communicate with backend services through secure APIs.

Application Layer

The Application Layer contains the business logic responsible for:

- Authentication
- AI recommendations
- Rewards calculations
- Watch Party management
- Creator Marketplace operations
- Payment processing
- Search functionality

Database Layer

The Database Layer stores all operational and analytical data.

It should include the STREAMORA relational database designed in Chapter Nine.

13.3 User Authentication and Account Management

Developers should implement a secure authentication system supporting:

- User registration
- Email verification
- Login
- Logout
- Password reset
- Password encryption
- Session management
- Two-Factor Authentication (2FA)

Additional Features

Users should also be able to:

- Update their profiles
- Upload profile pictures
- Change passwords
- Manage subscriptions

- Delete accounts if required

13.4 AI Recommendation Engine

One of STREAMORA's unique selling points is its AI-powered recommendation system.

The recommendation engine should use multiple data sources, including:

- Watch history
- User mood
- Search history
- Favorite genres
- Ratings
- Time of day
- Recently watched content

Recommendation Workflow

The AI engine should:

1. Collect user interaction data.
2. Analyze behavioral patterns.
3. Generate personalized recommendations.
4. Display recommendations on the Home Page.
5. Learn continuously from accepted and rejected recommendations.

The recommendation engine should improve over time using machine learning models.

13.5 Search System

The search functionality should go beyond exact keyword matching.

Developers should implement:

Natural Language Search

Example:

"Show me a funny action movie with a female lead."

The system should interpret the request and provide relevant recommendations.

Intelligent Auto-Complete

While users type, the system should suggest:

- Movies
- Actors
- Genres
- Creators

Search History Logging

Every search should be stored within the Search History dataset.

This information supports:

- AI learning
- Trend analysis
- Content acquisition

13.6 Watch Party Module

The Watch Party feature should support real-time interaction between users.

Developers should implement:

Live Chat

Participants should exchange messages during movie playback.

Voice Communication

Optional voice channels should allow participants to communicate naturally.

Emoji Reactions

Users should react instantly without interrupting playback.

Live Polls

Participants should vote on:

- Next movie
- Break time
- Party themes

Invitations

Users should share Watch Party invitations using:

- Links
- QR codes
- Social media

13.7 Creator Marketplace

The Creator Marketplace should provide independent filmmakers with professional publishing tools.

Developers should implement:

Creator Registration

Separate creator accounts should include:

- Identity verification
- Payment information
- Portfolio

Video Upload

Creators should upload:

- Videos
- Posters
- Descriptions
- Categories
- Pricing

Revenue Dashboard

Creators should monitor:

- Revenue
- Views
- Watch time
- Subscribers
- Tips

Subscription Management

Fans should subscribe directly to creators.

Tip Payments

Users should support creators using secure payment systems.

13.8 Rewards System

The Rewards System should automatically calculate user achievements.

Developers should implement logic for:

- Points
- Badges
- Challenges
- Leaderboards

Reward Activities

Points should be awarded for:

- Watching movies
- Writing reviews
- Inviting friends
- Joining Watch Parties
- Daily logins
- Completing challenges

The reward engine should update balances automatically after qualifying activities.

13.9 Database Management

The application should connect directly to the STREAMORA relational database.

Developers should ensure:

- Primary keys are enforced.
- Foreign keys maintain referential integrity.
- Indexes improve query performance.
- Transactions prevent inconsistent data.

Regular backups should be scheduled to protect against data loss.

13.10 API Development

A RESTful or Graph API should expose application services to the frontend.

Recommended API groups include:

User APIs

- Register user
- Login
- Update profile
- Subscription management

Movie APIs

- Get movie details
- Search movies
- Recommend movies
- Rate movies

Creator APIs

- Upload content
- View analytics
- Manage subscriptions
- Receive tips

Rewards APIs

- Retrieve points
- Update badges
- View challenges

Watch Party APIs

- Create party
- Join party
- Send reactions
- Submit polls

All APIs should include proper authentication, validation, and error handling.

13.11 Performance Optimization

To provide a smooth streaming experience, developers should implement:

- Lazy loading
- Content caching
- Image compression
- Video compression
- Database indexing
- CDN integration
- Efficient API responses

These optimizations reduce loading times and improve overall application performance.

13.12 Security Recommendations

Security should be incorporated throughout the platform.

Recommended measures include:

- HTTPS encryption
- Password hashing
- Role-Based Access Control (RBAC)
- Input validation
- SQL injection prevention
- Cross-Site Scripting (XSS) protection
- Cross-Site Request Forgery (CSRF) protection
- Secure payment processing
- Audit logging

Regular security testing should be conducted to identify vulnerabilities before deployment.

13.13 Scalability and Cloud Infrastructure

To support future growth, STREAMORA should be designed with scalability in mind.

Recommendations include:

- Cloud-based deployment
- Auto-scaling servers
- Load balancing
- Microservices architecture (where appropriate)
- Distributed databases
- Object storage for media files
- Queue services for background processing

This architecture ensures the platform can accommodate increasing numbers of users without compromising performance.

13.14 Monitoring and Analytics

Developers should implement monitoring tools to track system health and application performance.

Key metrics include:

- API response times
- Error rates
- Active users
- Server utilization
- Streaming quality
- Database performance

Application logs should be centralized to simplify troubleshooting and maintenance.

13.15 Integration with Power BI

The STREAMORA system should continuously feed operational data into the analytical environment.

Automated data refreshes should enable Power BI dashboards to display up-to-date information on:

- User engagement
- Content performance
- Creator revenue

- Rewards activity
- AI recommendation effectiveness
- Watch Party participation

This integration ensures decision-makers always have access to current business insights.

13.16 Expected Benefits

Implementing these technical recommendations will provide several benefits:

- Reliable and secure platform performance.
- Personalized user experiences through AI recommendations.
- Increased user engagement via Watch Parties and Rewards.
- Sustainable monetization opportunities for creators.
- Efficient database management and reporting.
- Scalability to support future growth.
- Seamless collaboration between frontend, backend, and analytics teams.

CHAPTER 14

FUTURE ENHANCEMENTS

14.1 Introduction

The entertainment streaming industry is constantly evolving due to rapid technological advancements, changing user expectations, and increasing competition. To remain competitive, STREAMORA should continuously improve its platform by introducing new technologies, expanding its services, and leveraging data-driven innovation.

Although the proposed platform already includes unique features such as AI-powered personalization, Watch Parties, a Creator Marketplace, and a Rewards System, there are additional opportunities that can further enhance the user experience and strengthen STREAMORA's market position.

This chapter outlines potential future enhancements that can be considered as the platform grows.

14.2 Advanced Artificial Intelligence

The current recommendation engine uses user behavior, viewing history, search history, mood, and preferences. In future versions, STREAMORA can implement more advanced AI capabilities.

Predictive Recommendations

Instead of recommending content only after users interact with the platform, AI can anticipate user preferences based on historical behavior and emerging trends.

For example, the system could suggest content before a user begins searching, reducing the time spent browsing.

Context-Aware Recommendations

Future AI models can consider additional contextual information such as:

- Weather conditions
- Holidays
- Special events
- Location (where appropriate and with user consent)
- Viewing companions (individual, family, or friends)

These factors can improve recommendation relevance.

14.3 Conversational AI Assistant

STREAMORA could introduce an AI-powered virtual assistant that interacts with users through natural language.

Example requests include:

- "Recommend a comedy for family movie night."
- "Find documentaries under 90 minutes."
- "Show movies similar to the last one I watched."
- "Create a weekend playlist."

The assistant could also answer account-related questions and help users navigate the platform.

14.4 Personalized User Profiles

Future versions of STREAMORA can offer deeper personalization.

Possible enhancements include:

- Personalized home page layouts.
- Custom content categories.
- Favorite actor and director tracking.
- Personalized notifications.
- Smart reminders for unfinished movies.
- Adaptive recommendations based on changing interests.

These features would improve user engagement and satisfaction.

14.5 Live Streaming Integration

STREAMORA can expand beyond on-demand entertainment by supporting live streaming.

Possible live content includes:

- Sports events.
- Music concerts.
- Award ceremonies.
- Live interviews.
- News programs.
- Educational events.

This would broaden the platform's audience and increase daily engagement.

14.6 Expanded Creator Marketplace

The Creator Marketplace can evolve into a comprehensive platform for independent content creators.

Future enhancements may include:

- Live broadcasts by creators.
- Premium creator memberships.
- Sponsorship opportunities.
- Merchandise sales.
- Collaborative projects between creators.
- Creator learning resources and analytics.

These additions can attract more creators and enrich the platform's content library.

14.7 Enhanced Rewards and Gamification

The Rewards System can be expanded to include more interactive and competitive features.

Potential additions include:

- Seasonal challenges.
- Leaderboards.
- Achievement levels.
- Daily login rewards.
- Community competitions.
- Referral milestones.
- Limited-time campaigns.

These features encourage regular platform usage and strengthen user loyalty.

14.8 Community Features

STREAMORA can develop a stronger sense of community by introducing social features.

Examples include:

- User discussion forums.
- Public movie reviews.
- Community playlists.
- Fan clubs for creators.
- User-generated watchlists.
- Movie recommendation sharing.
- Social media integration.

These features can increase user interaction and create stronger connections among viewers.

14.9 Advanced Analytics

As STREAMORA grows, the analytics platform should also evolve.

Future analytical capabilities may include:

- Predictive churn analysis.

- Customer lifetime value estimation.
- Recommendation performance optimization.
- Real-time engagement monitoring.
- Marketing campaign effectiveness.
- Revenue forecasting.
- Audience segmentation using machine learning.

These insights can support more effective business planning and product development.

14.10 Mobile and Smart Device Expansion

To reach a wider audience, STREAMORA should continue expanding its availability across devices.

Future support could include:

- Smart TVs.
- Gaming consoles.
- Streaming devices.
- Wearable technology.
- In-car entertainment systems.

Maintaining a consistent experience across platforms will improve accessibility and convenience.

14.11 Accessibility Improvements

Future updates should continue improving accessibility to ensure the platform is usable by everyone.

Potential enhancements include:

- Real-time subtitle translation.
- Sign language overlays for selected content.
- Audio descriptions for visually impaired users.
- Improved voice navigation.
- Enhanced keyboard shortcuts.
- Customizable accessibility profiles.

These improvements promote inclusivity and broaden the platform's reach.

14.12 Data Privacy and Security Enhancements

As the platform grows, protecting user information becomes increasingly important.

Future security initiatives should include:

- Continuous security monitoring.
- Multi-factor authentication.
- Advanced fraud detection.
- Encryption of sensitive user data.
- Regular security audits.
- Privacy dashboards allowing users to manage their personal data.

Strong security practices help build trust and ensure compliance with relevant data protection regulations.

14.13 International Expansion

STREAMORA can prepare for global growth by supporting multiple languages and regional content.

Future initiatives include:

- Multilingual user interfaces.
- Localized recommendations.
- Region-specific content catalogs.
- Multiple payment options.
- Partnerships with local creators and studios.
- Regional marketing campaigns.

International expansion will help STREAMORA reach new audiences and diversify its content offerings.

14.14 Emerging Technologies

To remain innovative, STREAMORA should monitor and evaluate emerging technologies that could enhance the platform.

Potential areas include:

- Augmented Reality (AR) experiences for promotional content.
- Virtual Reality (VR) movie viewing environments.
- AI-assisted content tagging and moderation.
- Intelligent video quality optimization.
- Blockchain-based royalty tracking for creators.
- Edge computing to improve streaming performance.

These technologies should be assessed based on user value, technical feasibility, and business objectives before implementation.

14.15 Strategic Roadmap

To guide future development, STREAMORA can adopt a phased implementation approach.

Short-Term (0–12 Months)

- Launch the core streaming platform.
- Implement AI recommendations.
- Introduce the Rewards System.
- Release Watch Party functionality.
- Launch the Creator Marketplace.

Medium-Term (1–3 Years)

- Expand AI capabilities.
- Introduce conversational AI.
- Improve creator analytics.
- Expand to additional devices.
- Launch community features.

Long-Term (3–5 Years)

- Support live streaming.
- Introduce AR/VR experiences where appropriate.
- Expand internationally.
- Implement predictive analytics.

- Continuously enhance accessibility and security.

14.16 Expected Benefits

Implementing these future enhancements is expected to:

- Increase user engagement and retention.
- Improve personalization and content discovery.
- Strengthen support for independent creators.
- Expand STREAMORA into new markets.
- Enhance platform security and accessibility.
- Enable more informed business decisions through advanced analytics.
- Maintain STREAMORA's competitiveness in the evolving streaming industry.

CHAPTER 15

CONCLUSION

15.1 Introduction

This project set out to explore how data analytics can support the design and development of a modern entertainment streaming platform named **STREAMORA**. The study began by analyzing an existing entertainment dataset to understand user behavior, content performance, and platform usage. It then identified the limitations of the available data and proposed additional datasets, database structures, dashboards, and business recommendations required to support STREAMORA's innovative features.

The project demonstrated that effective data analytics is not limited to reporting historical events but also plays a vital role in guiding product design, improving user experiences, and supporting strategic business decisions.

15.2 Summary of the Study

The project was completed in several phases, each contributing to the overall development of the STREAMORA platform.

The key activities carried out include:

- Reviewing and understanding the provided entertainment dataset.
- Assessing data quality through cleaning and validation.
- Conducting exploratory data analysis (EDA).
- Identifying gaps in the existing dataset.
- Designing additional datasets to support new platform features.
- Developing a relational database schema for STREAMORA.
- Building interactive Power BI dashboards.
- Generating business insights from the data.
- Producing recommendations for UI/UX designers.
- Providing technical recommendations for web developers.
- Identifying future opportunities for platform growth and innovation.

These activities created a comprehensive data-driven foundation for the development of STREAMORA.

15.3 Key Findings

The analysis produced several important findings that will influence the development of the platform.

Existing Dataset Limitations

The original entertainment dataset was useful for analyzing historical viewing behaviour but lacked information required to support advanced streaming features such as:

- AI-powered personalization.
- Mood-based recommendations.
- Creator monetization.
- Watch Parties.
- Rewards and gamification.
- Search behavior analysis.

These gaps highlighted the need for additional datasets.

User Behavior

The analysis showed that users value personalized experiences and are more likely to engage with platforms that recommend relevant content quickly and accurately.

Behavioral data such as watch history, ratings, search activity, and preferences provide valuable information for improving customer satisfaction.

Data as a Strategic Asset

The project demonstrated that data should not simply be collected but actively used to improve products and services.

Well-designed datasets allow organizations to:

- Understand customers.
- Predict future behavior.
- Improve recommendations.
- Increase engagement.
- Support business growth.

Importance of Data Visualization

The Power BI dashboards transformed complex datasets into interactive visual reports that enable decision-makers to identify trends and monitor key performance indicators efficiently.

These dashboards simplify communication between technical teams and business stakeholders.

15.4 Project Contributions

This project contributes to STREAMORA in several important ways.

Business Contribution

The research provides management with valuable insights that support strategic planning, customer engagement, and long-term platform growth.

Technical Contribution

The proposed relational database and data model provide developers with a structured foundation for implementing STREAMORA's core functionalities.

Design Contribution

The recommendations for UI/UX designers ensure that the interface is intuitive, user-centered, and aligned with customer expectations.

Analytical Contribution

The Power BI dashboards demonstrate how data analytics can be integrated into product development to support evidence-based decision-making.

Innovation Contribution

The introduction of AI-powered personalization, creator monetization, social viewing experiences, and gamification differentiates STREAMORA from many traditional streaming platforms.

15.5 Practical Implications

The findings of this project have practical implications for several stakeholder groups.

UI/UX Designers

Designers can create interfaces that are informed by real user needs rather than assumptions.

Web Developers

Developers have a clearly defined database structure and technical requirements that support efficient implementation.

15.6 Limitations of the Study

Although the project achieved its objectives, several limitations should be acknowledged.

- The research relied on a sample entertainment dataset, which may not fully represent the behavior of users on a live streaming platform.
- Some STREAMORA datasets were newly designed and therefore contain simulated data rather than production data.
- The proposed AI recommendation system was designed conceptually and was not trained or evaluated using live user interactions.
- The dashboards reflect the available sample data and may require refinement when integrated with real-time operational data.

Despite these limitations, the study provides a practical framework that can be expanded as STREAMORA grows.

15.7 Recommendations for Future Research

Future research can build upon this work in several areas.

Potential research topics include:

- Evaluating the performance of AI recommendation algorithms using real user data.
- Measuring the impact of Watch Parties on user engagement and retention.
- Studying the effectiveness of gamification strategies in streaming platforms.
- Investigating creator monetization models and revenue-sharing mechanisms.
- Exploring predictive analytics for customer churn and subscription renewal.
- Assessing the role of conversational AI in improving content discovery.

These studies would provide additional evidence to guide future platform enhancements.

15.8 Final Conclusion

The STREAMORA project demonstrates how data analytics can serve as the foundation for developing a modern, intelligent, and user-centered entertainment platform. Through the integration of data collection, cleaning, exploratory analysis, database design, dashboard development, and business intelligence, the project provides a comprehensive framework that supports informed decision-making across multiple disciplines.

The proposed datasets, relational database, Power BI dashboards, and strategic recommendations equip UI/UX designers and web developers with the information required to design and build a platform that prioritizes personalization, interactivity, creator empowerment, and user engagement.

As the streaming industry continues to evolve, organizations that effectively leverage data will be better positioned to understand customer needs, innovate rapidly, and remain competitive. STREAMORA has been conceptualized with this principle at its core, using data not only to describe user behavior but also to shape future experiences and drive continuous improvement.

By implementing the recommendations presented throughout this report, STREAMORA has the potential to establish itself as a distinctive, scalable, and data-driven entertainment platform that delivers value to users, creators, and stakeholders alike.

15.9 Closing Statement

This project demonstrates that successful digital products are built through collaboration between data analysts, UI/UX designers, and web developers. Data provides the evidence needed to understand users, validate ideas, and guide product decisions, while design and technology transform those insights into meaningful user experiences.

The STREAMORA project illustrates how a structured, data-driven approach can support the creation of an innovative streaming platform that combines intelligent personalization, social interaction, creator empowerment, and engaging user experiences. As the platform evolves, continuous analysis, user feedback, and iterative improvements will be essential to ensuring that STREAMORA remains relevant, competitive, and capable of meeting the changing needs of its users.